

Simple (Basic Syntax & Single Conditions)

1. Write an `if` statement to check if a given integer is positive.
2. Check if a string is empty using an `if-else` block.
3. Write a program that prints "Even" if a number is divisible by 2, and "Odd" otherwise.
4. Use an `if` statement to verify if a person's age is 18 or older to determine voting eligibility.
5. Write an `if-elif-else` block to check if a number is positive, negative, or zero.
6. Check if a specific character is a vowel using the `in` operator inside an `if` statement.
7. Write a script that checks if a list contains more than 5 elements.
8. Compare two numbers and print the larger one using an `if-else` statement.
9. Write an `if` statement that checks if a variable `x` is exactly equal to 10.
10. Check if a given string contains the substring "Python".
11. Print "Pass" if a student's score is 50 or above, otherwise print "Fail".
12. Use an `if` statement to check if a boolean variable `is_active` is `True`.
13. Write a program to check if a dictionary contains a specific key.
14. Create an `if-else` statement to check if a temperature is above 30 degrees (print "Hot") or not (print "Normal").
15. Write a condition to check if two strings are identical, taking case sensitivity into account.

Intermediate (Multiple Conditions, `elif`, & Basic Nesting)

16. Write a nested `if` statement to check if a number is positive, and if so, whether it is even or odd.
17. Use `if-elif-else` to categorize a student's grade: A (>90), B (80-90), C (70-79), D (60-69), F (<60).
18. Check if a given year is a leap year using logical `and` and `or` within an `if-else` block.
19. Write a nested `if` block to verify user login credentials: confirm the username exists first, then check the password.
20. Determine the largest of three different numbers using nested `if-else` statements.
21. Write an `if-elif-else` statement to determine the meteorological season based on the month (1-12).
22. Use `if` conditions to validate a password length (must be between 8 and 15 characters inclusive).
23. Check if a given triangle is equilateral, isosceles, or scalene based on its three side lengths using `if-elif-else`.
24. Write a nested condition checking if a user has a discount code; if valid, apply 20%, else 10%. If no code is entered, apply 0%.

25. Determine if a character is an uppercase letter, lowercase letter, or digit using `if-elif-else`.
26. Write an `if` statement that evaluates multiple combined conditions: `age > 21`, `has_id` is `True`, and `is_banned` is `False`.
27. Categorize Body Mass Index (BMI) into Underweight, Normal, Overweight, and Obese using an `elif` ladder.
28. Write a nested `if` that checks if a file path exists; if it does, check if the file is readable.
29. Check if a Cartesian coordinate (x, y) lies in the 1st, 2nd, 3rd, or 4th quadrant, or directly on an axis.
30. Create an `if-elif-else` structure to calculate a movie ticket price based on age (child, adult, senior) and day of the week (weekend surcharge).
31. Use a nested `if` to check if a string is a palindrome, but only execute the check if its length is greater than 3.
32. Write a script to calculate income tax based on three varying income brackets using `if-elif-else`.
33. Determine the type of a variable (`int`, `float`, `str`, `list`) using `if-elif-else` and the `isinstance()` function.
34. Check if an item is in a shopping cart; if yes, check if the quantity is greater than 5 to apply bulk pricing.
35. Write a nested conditional to simulate an ATM withdrawal: check if the PIN is correct, then check if the balance is sufficient, then dispense cash.

Advanced (Deep Nesting, Optimization & Complex Logic)

36. Refactor a deeply nested 4-level `if-else` block into a flat structure using early returns (guard clauses) within a function.
37. Write a single-line ternary conditional expression (`x if condition else y`) to replace a standard `if-else` block checking for even/odd numbers.
38. Use nested conditionals inside a list comprehension to filter strings from a list that start with 'A' and have a length greater than 5.
39. Combine `if-elif-else` logic with a dictionary mapping (dispatch table) to simulate a switch-case statement without using `match`.
40. Write a complex validation script that checks an email address for an @ symbol, a ., and a valid domain ending using nested `ifs` without importing the `re` module.
41. Implement Rock-Paper-Scissors game logic using an optimized `if-elif-else` structure that avoids redundant checks.
42. Evaluate short-circuiting: Write a condition where the second check (which would raise a `ZeroDivisionError`) is safely protected by the first check in a single `if` statement.

43. Write nested `if-else` statements to solve the FizzBuzz problem, then rewrite the exact same logic without using `elif` or `and`.
44. Implement a state machine using `if-elif-else` to handle transitions between 'START', 'PROCESSING', 'PAUSED', and 'STOPPED' states.
45. Use nested conditionals to parse a date string (YYYY-MM-DD) and validate the day based on the specific month and leap year rules.
46. Write a recursive function containing an `if-else` base case to calculate the Fibonacci sequence up to `n`.
47. Handle complex permissions: Write an `if` block checking if a user `is_admin`, OR if they `is_moderator` AND `owns_post`, OR if the post `is_public`.
48. Use the walrus operator (`:=`) within an `if-elif` block to assign the result of a regex match and evaluate it simultaneously.
49. Implement a single step of a binary search algorithm using nested `if-else` to adjust the high and low pointers based on the target value.
50. Refactor a complex `if-elif-else` chain that checks multiple type instances into Python 3.10+ `match-case` (structural pattern matching).